

PH 627: Topological Quantum Matter

Course Code	PH 627
Course Name	Topological Quantum Matter
L-T-P-C	3-0-0-3
Prerequisites	PH513, PH523
Intended for	UG/PG/I-PhD/PhD
Approval	52nd BoA (Nov 2023)

Course Contents

- The basics: Potentials in quantum mechanics, Aharonov-Bohm effect, Monopoles in physics, Berry phase. [4 Lectures]
- Symmetries: Time-reversal symmetry (TRS) in classical and quantum mechanics, TRS operator, Kramer's degeneracy, Symmetries in momentum space, Inversion symmetry, particle-hole symmetry, ten-fold classification. [6 Lectures]
- 1D Lattice models: Lattice models and band in momentum space, Peierl's instability, Su-Schrieffer-Heeger model, Berry phase effect on dynamics, topological index, charge fractionalization. [6 Lectures]
- Quantum Hall Effect: Kubo formula and TKNN invariant, quantization of Hall conductance, QHE in 2DEG, Landau levels, QHE in graphene. [6 Lectures]
- Topological insulators: Graphene, Dirac points, topological insulators, , Anomalous Hall Effect, BHZ model, edge states of BHZ model. Kane-Mele model, Z₂ invariant, 3D topological insulators, strong and weak TI. [6 Lectures]
- Topological metals: Accidental degeneracies, Weyl and Dirac fermions, symmetry analysis, chiral anomaly, anomalous Hall effect, Fermi arcs, Weyl semimetals, Dirac semimetals. [5 Lectures]
- Majorana fermions: Topological superconductivity, Majorana fermions, Majorana modes in chiral p-wave superconductors, Majorana modes in Rashba spin-orbit coupled semiconductors, detection of Majorana fermions. [5 Lectures]
- Miscellaneous Topics: Quantum Spin Liquid, RVB, Kitaev model, spin fractionalization. Materials research, experimental realization of several topological phases in TIs and topological metals. Electronic structure theory aspect of topological phases. [4 Lectures]

Text Books

1. B. Andrei Bernevig, Taylor L. Hughes, Topological Insulators and Topological Superconductors, Princeton University Press, 2013
2. Shun-Qing Shen, Topological Insulators: Dirac Equation in Condensed Matter, Springer, 2012

References

1. R. Shankar, **Topological Insulators - A review**, <https://arxiv.org/pdf/1804.06471.pdf>
2. M. Z. Hasan, C. L. Kane, **Topological Insulators**, <https://arxiv.org/pdf/1002.3895.pdf>
3. Xiao-Liang Qi and Shou-Cheng Zhang, **Topological insulators and superconductors**, <https://arxiv.org/pdf/1008.2026.pdf>
4. N.P. Armitage, E. J. Mele, Ashvin Vishwanath, **Weyl and Dirac Semimetals in Three-Dimensional Solids**, <https://arxiv.org/pdf/1705.01111.pdf>
5. David Vanderbilt, **Berry Phases in Electronic Structure Theory**, Cambridge University Press, 2018